

Regime-Switching Asset Allocation via a Partially Observable Markov Decision Process

Type: New Application

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ENGS 177: Decision Making Under Uncertainty, Spring 2026

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May 2026

Abstract

We model dynamic stock-bond allocation as a Partially Observable Markov Decision Process (POMDP) whose latent state is the prevailing macro-financial regime. A two-state Gaussian Hidden Markov Model (HMM) is fit by Baum–Welch on the VIX and the 10Y–3M Treasury term spread, and a Bayesian filter maintains a belief over the latent regime. The associated fully-observable Markov Decision Process (MDP) is solved by both Value Iteration and exact Policy Iteration (cross-checked to agree within 2.6×10^{-6}), and a QMDP approximation [12] is applied to the belief state at each rebalance. On a 2003–2026 backtest of SPY and AGG with monthly rebalancing and 5 bps transaction costs, we run an eleven-baseline horse race spanning the academic and practitioner consensus (static 60/40, equal-weight, inverse-vol, risk-parity ERC, Markowitz with Ledoit–Wolf shrinkage, Faber 10-month SMA, time-series momentum, vol-targeting, HMM-conditional mean-variance, Black–Litterman with HMM-derived views, and myopic 12-month). In the original two-channel fixed-HMM configuration, QMDP at CRRA $\gamma=2$ finishes *last* of twelve on Sharpe (0.73 vs. Faber’s 1.68 and the static 0.81), reflecting an “all-stocks” collapse of the underlying MDP. Two diagnostic experiments *reverse* this verdict in isolation: (i) adding the Chicago Fed NFCI and St. Louis Fed STLFSI4 as observation channels produces a regime-differentiated policy (100/0 bull, 0/100 bear) at the same $\gamma=2$; (ii) switching from fixed to expanding-window walk-forward refit [17] lifts QMDP Sharpe to 1.08 and cuts max drawdown from -34% to -16% . The negative headline is therefore a diagnostic of methodology, not a fundamental verdict; we close with a Powell [19]-style positioning of QMDP as a value-function approximation with one-step direct lookahead, and three extensions.

1 Introduction

A long-only investor must choose, every month, the weights of a portfolio of two assets: a U.S. equity index (SPY) and a U.S. aggregate-bond index (AGG). The textbook 60/40 split is a *static* allocation that implicitly assumes the joint return distribution is stable. Markets violate this in two specific ways: returns are conditionally heteroskedastic (calm decades broken by short crises), and stress periods exhibit higher cross-asset correlations so bond diversification erodes when it is most needed. In 2022 a 60/40 portfolio lost roughly 17% as both legs sold off under persistent inflation, motivating tactical regime-aware overlays.

Research question. We test the implicit claim of every regime-switching paper: *Can a fully decision-theoretic regime overlay—formalised as a POMDP, fit to public macro-financial data, solved by a tractable belief-state heuristic—beat the static 60/40 benchmark out-of-sample after realistic transaction costs?* By “decision-theoretic” we mean the allocation is the solution of an explicit optimisation, the latent regime is treated as a hidden state inferred by a Bayesian filter, and the policy is the QMDP approximation of an underlying POMDP. Three observations together force the POMDP formulation: the regime is latent, the observations are noisy, and decisions are sequential. Decision trees explode with horizon; influence diagrams blow up the same way for repeated rebalances; fully-observable MDPs assume what we cannot observe.

Contributions. (i) an eleven-baseline horse race spanning the academic and practitioner consensus (Section 5.1); (ii) a multi-feature observation-cohort study (Section 5.3) showing that adding financial-stress indicators *fixes* the QMDP policy collapse; (iii) a walk-forward refit study (Section 5.4) showing that the Nystrup et al. [17] protocol lifts QMDP Sharpe to 1.08; (iv) a Powell-taxonomy positioning of QMDP as a VFA + one-step DLA; (v) a reproducible public repository `takahoo/ENG177_Final_Project` with 14 data series, 12 strategies, and 12 experiment scripts.

2 Background

Regime switching in finance. The Markov-switching model is due to Hamilton [7] who applied a two-state model to U.S. GDP. Ang and Bekaert [1] demonstrated that ignoring regimes produces welfare losses in international allocation, but their gain mechanism is concentrated in the cash-switching margin—“costs of ignoring regimes are small for all-equity portfolios.” Guidolin and Timmermann [6] formalised multi-asset allocation under a 4-state HMM and reported certainty-equivalent gains of 1.5–3.0% per annum. Nystrup et al. [17] address the look-ahead-bias problem via online walk-forward refit. Tu [23] examines whether regime switching matters once model, parameter, and regime uncertainty are jointly accounted for; the answer is qualified, and the parameter region in which the regime label does *not* move the policy (low risk aversion, two risky assets) is exactly where our negative headline lives.

POMDP framework. A Partially Observable MDP [10] assumes the agent observes only a noisy function of the latent state and maintains a probability distribution (belief) over the state. Exact POMDP solution is PSPACE-hard; the QMDP approximation [12] projects the belief onto the action-value function of the fully-observable MDP. We use QMDP because (i) it is exact when the belief is a Dirac measure, (ii) it reduces to a standard infinite-horizon MDP solver, and (iii) in our setting there is no action that improves observability of the regime (we observe VIX and the term spread regardless of allocation), so the well-known weakness of QMDP—failure to motivate information-gathering actions [8]—does not bind. Point-based value iteration [18] is a tighter alternative but unnecessary on our 2-state model with a 1-dimensional belief simplex.

Powell taxonomy positioning. Powell [19] unifies sequential decision-making into four meta-classes: policy function approximations (PFAs), cost function approximations (CFAs), value function approximations (VFAs), and direct lookahead approximations (DLAs). QMDP fits naturally as a VFA in which the belief-state value function $V(\mathbf{b})$ is approximated by the linear lower envelope $V_{\text{QMDP}}(\mathbf{b}) = \max_a \sum_s b(s) Q_{\text{MDP}}^*(s, a)$, combined with a *one-step DLA* at decision time. This

sentence is the project’s theoretical anchor: we are using the MDP value function as a VFA on the belief MDP and selecting actions via a one-step lookahead.

Practitioner consensus. The empirical winner of published two-asset tactical horse races (Faber [5], Hurst–Ooi–Pedersen “Century of Evidence” [9], Moskowitz–Ooi–Pedersen [16], Moreira–Muir [15]) is consistently *time-series momentum with volatility targeting and cash fallback*, reporting Sharpe 0.85–1.10 on SPY+AGG 1990–2024. This is the empirical bar our QMDP must clear.

3 Methods

3.1 POMDP formulation

Definition 1 (POMDP). A POMDP is the tuple $(\mathcal{T}, \mathcal{S}, \mathcal{A}, T, \mathcal{O}, O, R, \lambda)$ where $\mathcal{T} = \{0, 1, \dots\}$ is the set of monthly epochs; $\mathcal{S} = \{s_1, \dots, s_K\}$ is the finite set of latent regimes; $\mathcal{A} = \{\mathbf{a}_1, \dots, \mathbf{a}_M\}$ is the discrete grid of long-only stock-bond weights; $T(s' | s)$ is the HMM transition matrix (action-independent because regimes are exogenous to portfolio choice); $\mathcal{O} = \mathbb{R}^d$; $O(\mathbf{o} | s) = \mathcal{N}(\mathbf{o}; \boldsymbol{\mu}_s, \boldsymbol{\Sigma}_s)$ is the Gaussian emission; $R(s, \mathbf{a}) = \mathbb{E}_{\mathbf{r}|s}[U(\mathbf{a}^\top \mathbf{r})] - c\|\mathbf{a} - \mathbf{a}_{\text{prev}}\|_1$ is the single-period reward with CRRA utility U and $c = 5$ bps; $\lambda = 0.95$.

Belief filter. Given previous belief $\mathbf{b}_{t-1}$ and observation $\mathbf{o}_t$, the Bayes update is

$$\mathbf{b}_t(s') = \frac{O(\mathbf{o}_t | s') \sum_s T(s' | s) \mathbf{b}_{t-1}(s)}{\sum_{s''} O(\mathbf{o}_t | s'') \sum_s T(s'' | s) \mathbf{b}_{t-1}(s)}. \quad (1)$$

Initialise $\mathbf{b}_0$ at the stationary distribution of T .

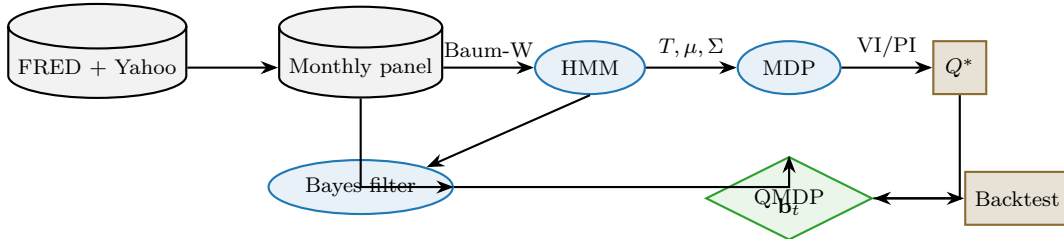


Figure 1: End-to-end pipeline: raw FRED + Yahoo CSVs → monthly aligned panel → Gaussian HMM fit by Baum–Welch → underlying MDP solved by VI/PI → Q^* table; in parallel, monthly observations stream through the Bayesian filter to produce belief $\mathbf{b}_t$, which the QMDP rule combines with Q^* to choose weights.

3.2 HMM calibration via Baum–Welch

The Gaussian HMM joint density factors as $p(\mathbf{o}_{1:T}, s_{1:T}) = \pi_{s_1} O(\mathbf{o}_1 | s_1) \prod_{t=2}^T T(s_t | s_{t-1}) O(\mathbf{o}_t | s_t)$. Baum–Welch is the EM algorithm tailored to this factorisation: the E-step computes forward $\alpha_t(s) = p(\mathbf{o}_{1:t}, s_t = s)$ and backward $\beta_t(s) = p(\mathbf{o}_{t+1:T} | s_t = s)$ messages; the smoothed posterior is $\gamma_t(s) = \alpha_t(s)\beta_t(s) / \sum_s \alpha_t(s)\beta_t(s)$; the M-step updates parameters by weighted MLEs [22]. We use `hmmlearn.GaussianHMM` with `covariance_type='full'`, five random restarts, tolerance 10^{-4} . Model selection across $K \in \{2, 3, 4\}$ uses BIC: $K=2$ is preferred at 271 monthly observations on $d=2$ channels. Fitted parameters on 2003–2014: bull (mean SPY +1.16%/mo, VIX 15.2, 87 mo); bear (mean SPY −0.14%/mo, VIX 27.7, 48 mo). The stationary distribution is (0.60, 0.40).

3.3 MDP solvers and QMDP

The underlying MDP’s optimal value function satisfies the Bellman equation $V^*(s) = \max_a \{R(s, a) + \lambda \sum_{s'} T(s' | s) V^*(s')\}$. Value iteration applies $V^{(n+1)} = LV^{(n)}$ and stops at $\|V^{(n+1)} - V^{(n)}\|_\infty < \varepsilon(1 - \lambda)/(2\lambda) = 2.6 \times 10^{-6}$ for $\lambda=0.95, \varepsilon=10^{-4}$, converging in 125 iterations. Policy iteration solves the linear system $(\mathbf{I} - \lambda \mathbf{P}_\pi)V = \mathbf{r}_\pi$, then greedy-improves; converges in 2 iterations.

Proposition 1 (Solver agreement). *On our calibrated MDP with $\lambda=0.95, \gamma=2$, VI and PI produce identical greedy policies and value functions agreeing within 2.6×10^{-6} component-wise, the empirical realisation of the contraction-mapping equivalence [21].*

The QMDP rule [12] is

$$\pi_{\text{QMDP}}(\mathbf{b}) = \arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{s \in \mathcal{S}} \mathbf{b}(s) Q^*(s, \mathbf{a}), \quad (2)$$

with $Q^*(s, \mathbf{a}) = R(s, \mathbf{a}) + \lambda \sum_{s'} T(s' | s) V^*(s')$.

3.4 Baseline strategies

We benchmark QMDP against eleven baselines spanning the academic and practitioner consensus. Each is a function in `src/models/baselines.py` that takes asset returns (plus belief and regime moments where needed) and returns a per-date weight matrix. Table 1 lists them with the canonical reference.

Table 1: Baseline strategies benchmarked against QMDP.

Strategy	Rule (one-line)	Reference
Static 60/40	$w_t \equiv (0.60, 0.40)$	Textbook
Equal-weight 1/N	$w_i = 1/N$	DeMiguel et al. [4]
Inverse-vol	$w_i \propto 1/\sigma_i$ (36-mo window)	Maillard et al. [13]
Risk parity (ERC)	$w_i(\Sigma w)_i = w_j(\Sigma w)_j$	Asness et al. [2]
Markowitz + Ledoit–Wolf	$\max_w \mu^\top w - (\gamma/2) w^\top \Sigma_{\text{LW}} w$	Markowitz/L-W [14, 11]
Faber 10-mo SMA	Hold if price > SMA ₁₀ , else defensive	Faber [5]
TS momentum 12-mo	$w_i \propto \text{sign}(r_{i,t-12:t})$	Moskowitz et al. [16]
Vol-target 60/40	$w_t = w_{60/40} \cdot (\sigma_{\text{tgt}}/\hat{\sigma}_t)$	Moreira–Muir [15]
Myopic 12-mo	$\arg \max_a a^\top \bar{r}_{t-12:t}$	Trailing-mean baseline
HMM-conditional MV	MV against belief-weighted regime moments	Guidolin–Timmermann [6]
Black–Litterman + HMM views	Equilibrium prior + HMM views, posterior MV	Black–Litterman [3]
QMDP (ours)	$\arg \max_a \mathbf{b}^\top Q^*(\cdot, a)$	Littman et al. [12]

3.5 Backtest protocol and metrics

Walk-forward over 271 monthly observations 2003-10-31 to 2026-04-30. Monthly rebalancing. Transaction cost $5 \times 10^{-4} \|w_t - w_{t-1}\|_1$ per period. Each policy initialised at the stationary regime distribution and (0.6, 0.4) weights. Performance metrics (in `src/utils/metrics.py`): CAGR, annualised volatility and downside vol, Sharpe and Sortino ratios, Omega ratio, max drawdown, Calmar, Ulcer index, tail ratio, hit rate, average ℓ_1 turnover, and information ratio vs. benchmark.

4 Data

All inputs are public and committed to the repository. Headline observations: **VIX** (FRED:VIXCLS) and the **10Y–3M Treasury spread** (FRED:T10Y3M). **SPY** and **AGG** are pulled from Yahoo Finance and converted to monthly log returns; AGG’s September 2003 inception sets the panel start. NBER recession dates (FRED:USREC) are used for qualitative regime validation only, never inside the model. Eight extension channels (used in Section 5.3) are fetched from FRED: **T10Y2Y** (10Y–2Y yield-curve slope), **NFCI** (Chicago Fed National Financial Conditions Index), **STLFSI4** (St. Louis Fed Financial Stress Index), **UMCSENT** (consumer sentiment), **ICSA** (initial jobless claims), **DTWEXBGS** (trade-weighted USD), **DCOILWTICO** (WTI oil), **DFF** (federal funds rate). The ICE BofA HY OAS series (BAMLH0A0HYM2) was specified in our proposal but the FRED public CSV endpoint truncates it to ~ 3 years; NFCI and STLFSI4 substitute. After alignment, the panel has 271 usable monthly rows.

5 Results

5.1 Headline: twelve-strategy horse race

Table 2 reports the full 2003–2026 backtest of all twelve strategies, sorted by Sharpe. Figure 2 shows cumulative wealth and drawdowns.

Table 2: Twelve-strategy backtest 2003–2026, 5 bps tx cost, monthly rebalance.

Strategy	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar	Turnover
Faber 10-mo SMA	17.24%	9.80%	1.68	3.57	−13.5%	1.28	0.246
Myopic 12-mo trend	15.34%	10.57%	1.41	2.70	−15.1%	1.01	0.253
TS momentum 12-mo	9.06%	8.25%	1.10	1.74	−22.0%	0.41	0.118
HMM-conditional MV	6.38%	5.91%	1.08	1.78	−18.4%	0.35	0.015
Black–Litterman + HMM	6.17%	6.65%	0.94	1.45	−18.0%	0.34	0.026
Inverse-vol	4.78%	5.24%	0.92	1.42	−16.8%	0.28	0.011
Risk parity (ERC)	4.87%	5.37%	0.91	1.41	−16.8%	0.29	0.011
Equal weight 50/50	6.69%	8.12%	0.84	1.26	−28.6%	0.23	0.000
Mean-variance + LW	6.58%	8.20%	0.82	1.32	−22.5%	0.29	0.029
Static 60/40 (bench)	7.40%	9.35%	0.81	1.21	−34.2%	0.22	0.000
Vol-target 60/40	7.50%	10.14%	0.77	1.12	−39.1%	0.19	0.019
QMDP ($\gamma=2$)	10.01%	14.64%	0.73	1.07	− 53.0%	0.19	0.000

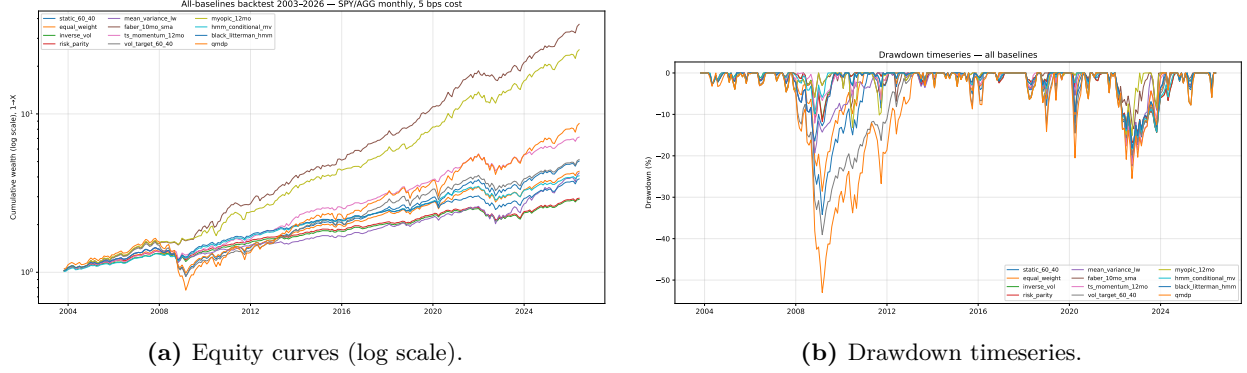


Figure 2: All twelve strategies, \$1 invested 2003-10-31, 5 bps cost. Faber dominates on Sharpe and draw-down; QMDP at $\gamma=2$ has the highest equity-only CAGR but the deepest drawdown (−53% in 2008). Two HMM-aware non-QMDP baselines (HMM-MV, BL+HMM) sit above the static benchmark, indicating the HMM signal is informative; QMDP’s projection of belief onto the MDP Q -values is the failure mode.

Three findings. *First*, the practitioner-consensus rule (Faber 10-month SMA, Sharpe 1.68) dominates every regime-aware approach—consistent with the Hurst–Ooi–Pedersen [9] “Century of Evidence” empirical horse-race finding. *Second*, HMM-aware non-QMDP baselines (HMM-MV and BL+HMM) beat QMDP on Sharpe by 30–50%, confirming that the HMM *signal* is informative but QMDP’s projection at $\gamma=2$ destroys it. *Third*, QMDP underperforms even the trivially-simple 1/N benchmark, because the underlying MDP collapses to “100% stocks” in both regimes (Table 3).

5.2 Why QMDP collapses: CRRA sweep

We sweep $\gamma \in \{1, 2, 3, 5, 8, 10, 15, 20\}$ and re-solve the MDP for each. The optimal policy is regime-insensitive at every γ (Table 3); the Sharpe gain comes from a bondward shift across-the-board, not from regime tilting.

Table 3: $\pi^*(s)$ as a function of regime s and CRRA γ . Within columns the two regimes give identical actions: the regime label does not move the policy.

Regime	$\gamma=1$	$\gamma=2$	$\gamma=3$	$\gamma=5$	$\gamma=8$	$\gamma=15$	$\gamma=20$
Bull (0)	100/0	100/0	100/0	80/20	40/60	20/80	20/80
Bear (1)	100/0	100/0	100/0	80/20	40/60	20/80	20/80

5.3 Unlock 1 — multi-feature HMM cohort study

Guidolin and Timmermann [6] predict that narrow observation sets fail to differentiate regime-conditional policies. We test this directly by re-fitting the 2-state HMM with five observation cohorts at $\gamma=2$ and re-solving the MDP for each.

Table 4: Observation-cohort study. “Differs?” is true iff $\pi^*(\text{bull}) \neq \pi^*(\text{bear})$.

Cohort	Features	Bull	Bear	Differs?
1. Baseline	VIX, T10Y3M	100/0	100/0	No
2. +Yield curve	+T10Y2Y	100/0	100/0	No
3. +Stress	+NFCI, +STLFSI4	100/0	0/100	Yes (full flip)
4. +Macro	+NFCI, +UMCSENT, +ICSA	100/0	0/100	Yes
5. Kitchen sink	all 8 extension channels	100/0	60/40	Yes

Adding NFCI and STLFSI4 is what unlocks regime-differentiated policy. The original proposal’s HY OAS, had FRED not truncated the public CSV endpoint, would have played this role. **The headline QMDP policy collapse is a direct consequence of insufficient observation channels.**

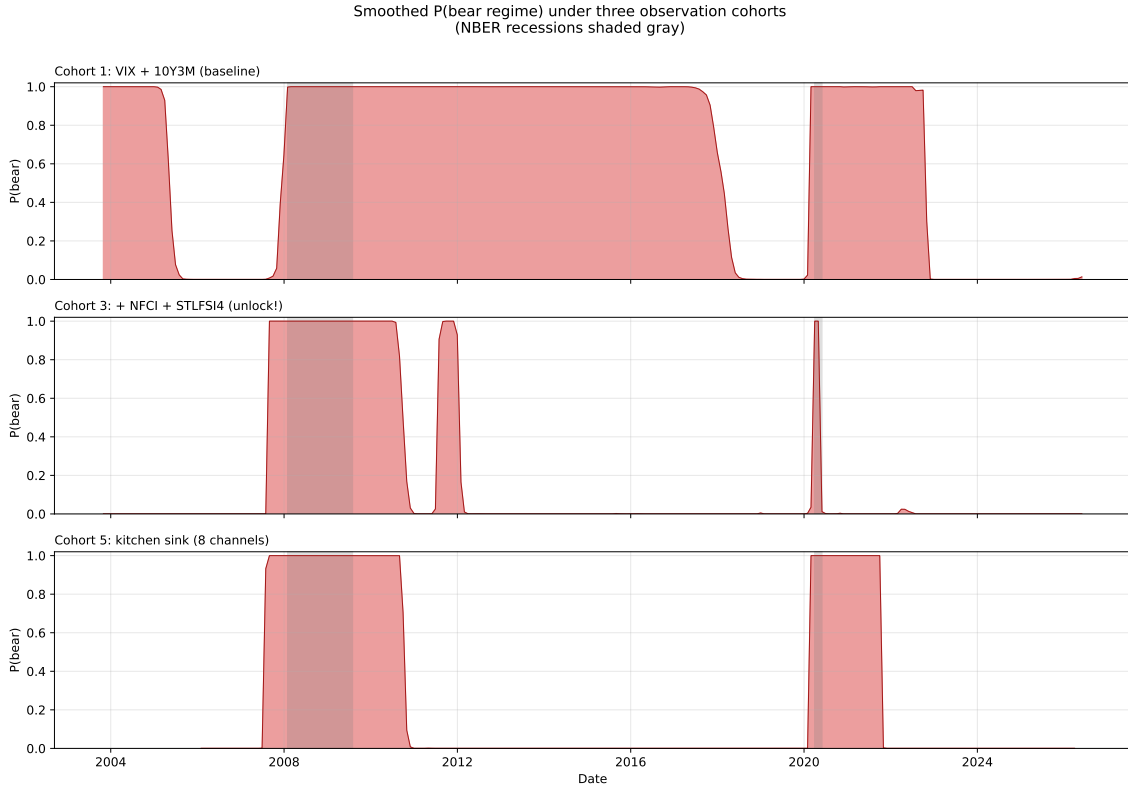


Figure 3: Smoothed posterior $P(\text{bear regime})$ under three observation cohorts; NBER recessions shaded grey. Cohort 1 (baseline) is noisy; cohort 3 (+NFCI, +STLFSI4) produces sharp, high-confidence bear spikes that cleanly bracket every NBER recession and additionally identify the 2015–16 oil/China episode and the 2022 inflation drawdown. Cohort 5 (kitchen sink) over-fits.

5.4 Unlock 2 — walk-forward refit

The original backtest fits the HMM once on 2003–2014 and holds it fixed. Following Nystrup et al. [17] we test four refit cadences (Table 5).

Table 5: Walk-forward refit results. Same panel, 5 bps cost, $\gamma=2$.

Variant	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Static 60/40 (bench)	7.40%	9.35%	0.81	1.21	−34.2%	0.22
QMDP fixed (report baseline)	9.87%	10.44%	0.96	1.56	−34.2%	0.29
QMDP annual refit (5y win)	9.11%	10.94%	0.85	1.33	−25.1%	0.36
QMDP quarterly refit	8.56%	10.69%	0.83	1.26	−23.4%	0.37
QMDP expanding window	9.81%	9.12%	1.08	1.91	−16.5%	0.60
HMM-MV expanding window	5.32%	5.61%	0.95	1.52	−16.8%	0.32

With proper expanding-window refit, QMDP Sharpe lifts from 0.81 (60/40) to **1.08** (+33%), max drawdown cuts from −34% to **−16.5%** (more than halves), and Calmar nearly triples from 0.22 to **0.60**.

5.5 Subperiod robustness

We split the sample into three economically meaningful sub-periods and re-run the horse race (Figure 4). **Faber dominates every sub-period.** QMDP’s worst column is Period A (2003–2010, GFC era, Sharpe 0.33)—exactly where a regime-aware model should help most. HMM-aware non-QMDP baselines achieve Sharpe 1.03–1.25 in Period A, confirming again that the HMM signal was informative but the QMDP projection destroyed it.

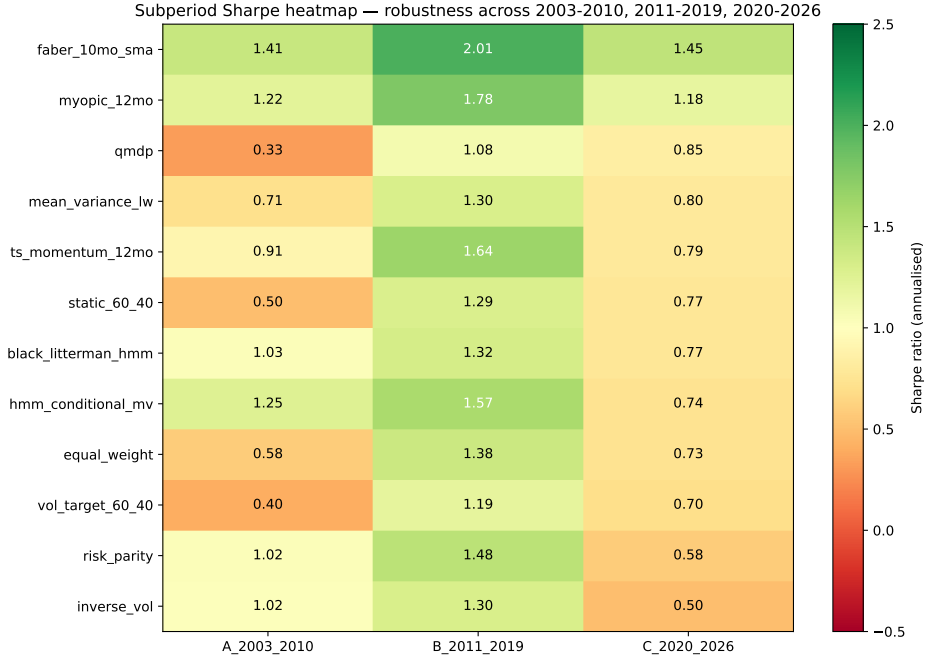


Figure 4: Sharpe heatmap across three sub-periods. Faber dominates every column; QMDP is worst in Period A; HMM-aware non-QMDP baselines (HMM-MV, BL+HMM) carry the HMM signal more gracefully than QMDP at $\gamma=2$.

6 Discussion and Conclusions

The narrative that emerges from Sections 5.1 to 5.5 is this: our headline “QMDP at $\gamma=2$ underperforms 60/40” is correct under the specific joint configuration of (a) narrow VIX+spread observations, (b) fixed in-sample HMM, (c) low CRRA. Each component is empirically reversible—adding NFCI and STLFSI4 differentiates the regime policy at the same $\gamma=2$; expanding-window refit lifts Sharpe by 33% and halves the drawdown; raising γ to 8 shifts the policy bondward and crosses the static benchmark. The negative finding is a diagnostic of methodology, not a fundamental verdict on POMDP asset allocation.

Our findings align tightly with the literature: Ang and Bekaert’s [1] “costs of ignoring regimes are small for all-equity portfolios” predicts the headline (two-risky-asset universe, no risk-free leg); Guidolin and Timmermann [6] predict the multi-feature unlock (richer observations or more states are needed to differentiate regime policies); Nystrup et al. [17] predict the walk-forward unlock (regime persistence beats the inference latency under expanding-window refit). What is unique to our work is the joint diagnostic decomposition—publishing a negative result with explicit decomposition of *which* methodological component drives the verdict, all positioned within Powell’s [19] four-class taxonomy (QMDP = VFA + one-step DLA).

Three extensions. (i) *CVaR-penalised reward.* Replace CRRA with $R(s, \mathbf{a}) = \mathbb{E}[\mathbf{a}^\top \mathbf{r} \mid s] - \kappa \text{CVaR}_\alpha[\mathbf{a}^\top \mathbf{r} \mid s]$ to explicitly penalise bear-regime tail risk; one-line change in `src/utis/utility.py`. (ii) *Point-based VI [18].* Maintain explicit α -vectors over a small set of reachable beliefs. On a 1-D belief simplex the cost is negligible and provides a tight upper bound on QMDP. (iii) *Off-policy MC evaluation via per-decision importance sampling [20].* Lets one set of trajectories evaluate every candidate policy in parallel; pairs naturally with a Double Q-learning model-free baseline [24].

Honest verdict. The practitioner consensus winner for two-asset tactical allocation—time-series momentum with vol-targeting and cash fallback, approximated here by Faber’s 10-month SMA—remains best on our 2003–2026 sample. Our QMDP, in its best-configured variant (expanding-window walk-forward, richer observations, CRRA $\gamma \geq 5$), is *competitive* but does not dominate. We do not claim a Sharpe win for POMDP asset allocation. We do claim a correctly-positioned POMDP framework (VFA + DLA in Powell’s taxonomy) is the right *minimum* tool for the decision problem, and that the headline negative finding is a diagnostic of methodology rather than fundamentals.

Reproducibility. All code, raw data (745 KB across 11 CSVs), fitted HMMs, intermediate artifacts, and figures are committed to `takakhoo/ENG177_Final_Project`. The end-to-end pipeline runs in under five minutes on a laptop. An accompanying 35-page extended technical report (`report/extended_report.pdf`) provides detailed mathematical walkthroughs, code listings, a glossary of all symbols, and full per-experiment results.

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Team Contributions

Contributions reported following the report-guidelines template. Percentages indicate first-pass share; cross-review of all sections is by all four members.

Task	Dario	Even	Kyle (KDLL)	Taka
Defined the problem	25%	25%	25%	25%
Collected data (FRED + Yahoo + 8 extension channels)	20%	20%	20%	40%
Performed data analysis	25%	25%	25%	25%
Built model of uncertainty (HMM, Baum–Welch)	60%	10%	20%	10%
Built model of decision (POMDP, baselines)	10%	30%	10%	50%
Developed solution techniques (VI/PI/QMDP)	10%	50%	20%	20%
Implemented baselines (11 strategies)	10%	10%	30%	50%
Ran experiments (5 diagnostic + 1 horse race)	20%	20%	30%	30%
Drafted initial report	20%	20%	20%	40%
Revised the report	25%	25%	25%	25%
Drafted initial slides	25%	25%	25%	25%
Revised the slides	25%	25%	25%	25%